Robotics Programming Lab

Research Metric Dictionary

Version 1.0 - comprehensive candidate metrics

Purpose: give the backend, instructor dashboards, research exports, SQA, and thesis analysis one consistent definition for every metric under consideration.

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Status: Proposed design specification. No application or database implementation is implied by this document.

Research question supported

Does test-driven evaluation with AI tutoring improve student learning, debugging behavior, efficiency, and experience compared with test-driven evaluation alone?

Scope and interpretation

This dictionary is deliberately broader than the first implementation. Core metrics are recommended for early collection; Secondary metrics add useful interpretation; Later metrics require more data, manual review, stronger study design, or advanced statistical analysis. Raw evidence must be preserved so derived metrics can be recalculated without rewriting history.

1. Standard definitions

These terms must be frozen before collecting real study data.

Term	Formal definition
Assigned student	A student connected to a study and assigned an assignment.
Started assignment	The student produced a meaningful event: code edit, test run, hint request, AI question, or submission.
Valid attempt	A grading request completed with a student-related result.
Invalid attempt	A duplicate, unauthorized, malformed, or infrastructure-failed request that never produced valid grading evidence.
Successful attempt	A valid attempt that satisfies the versioned assignment completion rule.
Completed assignment	The student produced at least one successful attempt.
Final attempt	The last valid attempt before the study deadline or approved extension.
Best attempt	The valid attempt with the highest score; the earlier attempt wins a tie.
Meaningful revision	Code changed beyond whitespace-only or comment-only edits.
Active time	Time between meaningful events, excluding event gaps longer than the configured idle threshold.
Recommended idle threshold	15 minutes, configurable and frozen at the study level.
Infrastructure failure	A grader, network, database, Docker, AI provider, or server failure not caused by student code.
Student error	A syntax, runtime, failed-test, timeout, unsafe-operation, output-limit, or size-limit result caused by submitted code.
AI interaction	A valid student question or hint request with a recorded tutor outcome.
Score percentage	100 x earned_score / maximum_score.
Study condition	The instructor-assigned experimental group, for example TDD_ONLY or TDD_AI.
Analysis population	Students satisfying the study's predefined eligibility and analysis rules.

2. Proposed canonical source entities

These are proposed canonical names, not confirmation of the current schema. Implementation must map them to versioned contracts.

Term	Formal definition
users	user_id, role, created_at, status
studies	study_id, title, starts_at, ends_at, idle_threshold_minutes
study_participants	study_id, user_id, participant_code, condition_id, assigned_at, consent_status
study_assignments	study_id, assignment_id, assignment_version, available_at, due_at
assignments	assignment_id, assignment_version, max_score, completion_rule
learning_objectives	objective_id, assignment_id, assignment_version
sessions	session_id, user_id, study_id, started_at, ended_at, active_seconds
attempts	attempt_id, session_id, user_id, assignment_id, assignment_version, condition_id, sequence_number, submitted_at, score, max_score, execution_status, code_hash, code_size_bytes
test_results	attempt_id, test_id, visibility, objective_id, outcome, execution_ms
error_events	attempt_id, error_category, error_fingerprint, student_caused
code_changes	attempt_id, previous_attempt_id, lines_added, lines_removed, meaningful_change
hint_interactions	interaction_id, attempt_id, hint_level, requested_at, delivered_at, status
ai_interactions	interaction_id, user_id, assignment_id, before_attempt_id, response_type, requested_at, responded_at, status
ai_reviews	interaction_id, correctness_rating, relevance_rating, leakage_flag, reviewer_id
assessments	user_id, study_id, assessment_type, score, max_score
surveys	user_id, study_id, assignment_id, question_id, response_value
system_events	request_id, service, event_type, status, duration_ms, occurred_at

3. Formal metric dictionary

Dashboard labels: Overview = whole study/class; Student = individual timeline; Assignment = content analysis; AI QA = tutor review; Research = comparison/export; Operations = system reliability; Data QA = research-data integrity.

A. Study population and participation

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
POP-01 Assigned students	Count distinct assigned user_id	study_participants.user_id, study_id	Active student assignments	Instructors and deleted test accounts	Overview	Core
POP-02 Eligible students	Count students satisfying eligibility rules	participant, consent, and study fields	Meets predefined protocol	Withdrawn, ineligible, or missing consent	Research	Core
POP-03 Students started	Distinct students with at least one meaningful event	sessions, attempts, hints, AI interactions	Events during study window	Login-only and system-generated events	Overview	Core
POP-04 Participation rate	100 x started / assigned	POP-01 and POP-03	Students assigned before deadline	Protocol-defined late additions	Overview	Core
POP-05 Students completed	Distinct students satisfying completion rule	attempts and completion_rule	Valid successful attempt	Infrastructure-only activity	Overview	Core
POP-06 Completion rate	100 x completed / started	POP-03 and POP-05	Started students	Never-started students	Overview	Core
POP-07 Assignment uptake rate	100 x started assignment / assigned assignment	study_assignments and activity	Assignment available to student	Instructor-disabled assignments	Assignment	Core
POP-08 Return-session rate	100 x students with 2+ sessions / students started	sessions.session_id, user_id	Separate valid activity periods	Reconnects and duplicate sessions	Overview	Secondary
POP-09 Study retention rate	100 x students reaching final activity / eligible	participant and activity data	Protocol-defined final activity	Withdrawals reported separately	Research	Secondary
POP-10 Withdrawal rate	100 x withdrawn / enrolled	consent_status and participant status	Enrolled participants	Administrative test accounts	Research	Core

B. Assignment performance and outcomes

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
OUT-01 First-attempt score	100 x first valid score / max score	attempt sequence, score, max_score	First valid student attempt	Infrastructure failures and duplicates	Student, Research	Core
OUT-02 Final-attempt score	Score percentage of last valid attempt	submitted_at, score, max_score	Last valid attempt before deadline	Attempts after deadline	Student, Research	Core
OUT-03 Best-attempt score	Maximum valid score percentage	attempt scores	Valid attempts in study window	Invalid attempts	Student, Assignment	Core
OUT-04 Absolute score gain	Final score% - first score%	OUT-01 and OUT-02	At least two valid attempts	Single-attempt students reported separately	Student, Research	Core
OUT-05 Best score gain	Best score% - first score%	OUT-01 and OUT-03	At least one valid attempt	Invalid attempts	Student	Secondary
OUT-06 Normalized gain	(final - first) / (100 - first)	first and final percentages	First score below 100	First score of 100	Research	Secondary
OUT-07 Successful solution	1 if completion rule is satisfied; otherwise 0	attempt outcome and completion_rule	Valid attempts	Infrastructure failures	Student, Research	Core
OUT-08 First-attempt success rate	100 x successful first attempts / valid first attempts	first attempt outcome	One first valid attempt per student	Students without valid attempts	Assignment	Core
OUT-09 Overall success rate	100 x successful students / students started	completion and participation	Started students	Never-started students	Overview, Assignment	Core
OUT-10 Test pass percentage	100 x passed tests / executed tests	test_results.outcome	Successfully executed tests	Infrastructure-skipped tests	Student, Assignment	Core
OUT-11 Public-test pass percentage	100 x passed public tests / executed public tests	visibility and outcome	Public tests	Hidden tests	Student, Assignment	Core
OUT-12 Hidden-test pass percentage	100 x passed hidden tests / executed hidden tests	visibility and outcome	Hidden tests	Public tests	Instructor, Research	Core
OUT-13 Objective mastery percentage	100 x mastered objectives / assessed objectives	objective mapping and outcomes	Objectives with sufficient mapped tests	Objectives without valid evidence	Student, Assignment	Core
OUT-14 Rubric score percentage	100 x earned rubric points / available points	rubric evidence	Published rubric version	Unassessed criteria	Instructor, Research	Secondary
OUT-15 Assignment completion status	Not started, started, partial, or completed	activity and completion_rule	Assigned students	Missing evidence marked unknown	Student, Overview	Core

C. Attempts and revision behavior

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
ATT-01 Total valid attempts	Count valid attempts	attempts.execution_status	Completed student-related results	Infrastructure failures and duplicates	Student, Assignment	Core
ATT-02 Attempts to first success	Sequence number of first successful attempt	sequence_number and outcome	Students who succeed	Non-completers are censored	Student, Research	Core
ATT-03 Unsuccessful attempts	Count valid attempts before success or deadline	attempt outcome	Valid student attempts	Infrastructure failures	Student, Assignment	Core
ATT-04 Meaningful revisions	Count attempts with meaningful_change=true	code_changes	Code-changing attempts	Whitespace/comment-only edits	Student	Core
ATT-05 Meaningful revision rate	100 x meaningful revisions / comparable attempts	code-change records	Attempts with previous snapshot	First attempt and missing snapshots	Student, Research	Secondary
ATT-06 No-change rerun rate	100 x identical-code attempts / reruns	attempts.code_hash	Consecutive reruns	First attempt	Student, Assignment	Secondary
ATT-07 Average code-change size	Mean(lines_added + lines_removed)	code_changes line counts	Meaningful revisions	Formatting-only changes	Student	Later
ATT-08 Median code-change size	Median changed lines	code_changes line counts	Meaningful revisions	Non-comparable snapshots	Research	Later
ATT-09 Progressing-attempt rate	Attempts increasing passed tests / comparable attempts	consecutive test results	Same assignment version	Infrastructure failures	Student, Research	Core
ATT-10 Regressing-attempt rate	Attempts losing passing tests / comparable attempts	consecutive test results	Same tests and assignment version	Changed test packages	Student, Assignment	Core
ATT-11 Stagnant-attempt rate	Attempts with no score/test improvement / comparable attempts	consecutive attempts	Same assignment version	Infrastructure failures	Student, Assignment	Secondary
ATT-12 Largest single-attempt improvement	Maximum consecutive score increase	attempt scores	Comparable valid attempts	Assignment-version changes	Student	Later

D. Time and engagement

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
TIME-01 Active working time	Sum event gaps at or below idle threshold	meaningful event timestamps	Events during assignment window	Long idle gaps	Student, Research	Core
TIME-02 Time to first attempt	First attempt time - first meaningful event	session and attempt timestamps	Students with valid attempt	Login-only activity	Student	Secondary
TIME-03 Time to first success	Active time from start to first success	events and success time	Successful students	Idle time and infrastructure downtime	Student, Research	Core
TIME-04 Time to final attempt	Active time through final valid attempt	event timestamps	Students with valid attempt	Idle gaps	Student	Secondary
TIME-05 Median time between attempts	Median active gap between valid attempts	attempt timestamps	Consecutive valid attempts	Idle gaps and system failures	Student, Assignment	Core
TIME-06 Time before first hint	Active time from start to first hint	start and hint timestamps	Hint users	Students without hints	Student, Research	Core
TIME-07 Time before first AI use	Active time from start to first AI interaction	start and AI timestamps	AI users	AI-unavailable students	Research	Core
TIME-08 Feedback-to-next-attempt time	Next attempt time - feedback delivery time	interaction and attempt timestamps	Feedback followed by attempt	No later attempt	Student, AI QA	Secondary
TIME-09 Session count	Count valid sessions	sessions.session_id	Sessions with meaningful activity	Login-only and reconnect sessions	Student	Core
TIME-10 Median session duration	Median active_seconds	sessions.active_seconds	Valid active sessions	Empty sessions	Student, Overview	Secondary
TIME-11 Abandonment duration	Active time before unfinished session ends	session events and completion	Unfinished sessions	Known system-forced shutdowns	Assignment	Later
TIME-12 Deadline lateness	Final submission time - due_at	submitted_at and due_at	Late attempts when allowed	Approved extensions	Instructor	Later

E. Errors, debugging, and TDD behavior

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
ERR-01 Syntax-error count	Count syntax-error attempts	error_category	Student-caused syntax errors	Platform parser failures	Student, Assignment	Core
ERR-02 Runtime-error count	Count student runtime errors	error_category	Valid executions	Container/platform failures	Student, Assignment	Core
ERR-03 Timeout count	Count student-code timeouts	execution_status	Confirmed submission timeout	Grader startup timeout	Student, Assignment	Core
ERR-04 Failed-test count	Sum failed tests across valid attempts	test_results.outcome	Executed tests	Skipped/unavailable tests	Student, Assignment	Core
ERR-05 Unsafe-code rejection count	Count security-policy rejections	execution status and error category	Confirmed policy violation	Sandbox infrastructure errors	Instructor, Operations	Core
ERR-06 Excessive-output count	Count output-limit terminations	execution_status	Student output exceeded limit	Logging duplication	Instructor	Core
ERR-07 Oversized-submission count	Count size-limit rejections	code_size_bytes and status	Submission exceeded limit	Transport corruption	Instructor	Core
ERR-08 Unique error categories	Count distinct student error categories	error_category	Student-caused errors	Infrastructure errors	Student, Assignment	Secondary
ERR-09 Repeated-error rate	Consecutive same-fingerprint errors / comparable error attempts	error_fingerprint and sequence	Same assignment version	Infrastructure errors	Student, Research	Core
ERR-10 Error persistence length	Maximum consecutive attempts with same fingerprint	error fingerprint	Comparable attempts	Version changes	Student, Assignment	Secondary
ERR-11 Error-resolution time	Active time from first appearance to later absence	error and event timestamps	Error followed by comparable attempt	Unresolved cases are censored	Student, Research	Core
ERR-12 Error-resolution attempts	Attempts from error appearance to resolution	attempt sequence	Resolved errors	Unresolved cases reported separately	Student, Assignment	Core
ERR-13 Regression count	Previously passing test becomes failing	consecutive test outcomes	Same test and version	Test-suite changes	Student, Assignment	Core
ERR-14 Regression recovery rate	Recovered regressions / total regressions	test history	Trackable regressions	Unresolved regressions	Student	Secondary
ERR-15 Failed-test response rate	Next revision addresses 1+ failed test / eligible revisions	test history and code changes	Meaningful consecutive revisions	No-change reruns	Research	Core
ERR-16 Error-free attempt rate	Attempts without student errors / valid attempts	attempt and error status	Valid attempts	Infrastructure failures	Assignment, Research	Secondary

F. Misconceptions and learning objectives

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
LRN-01 Misconception occurrence count	Count detected misconception evidence	instructor misconception rules	Evidence meeting published rule	Unreviewed AI guesses	Assignment	Core
LRN-02 Students affected by misconception	Distinct students with misconception evidence	misconception and user IDs	Valid evidence	Duplicate evidence in same attempt	Assignment	Core
LRN-03 Misconception prevalence	100 x affected students / students attempting	LRN-02 and participation	Students with valid attempts	Never-started students	Assignment, Research	Core
LRN-04 Misconception recovery rate	Students resolving misconception / students showing it	attempt history	Later comparable evidence exists	Unresolved cases reported separately	Research	Core
LRN-05 Attempts to misconception recovery	Attempts from first detection to resolution	attempt sequence	Resolved misconceptions	Version changes	Student	Secondary
LRN-06 Objective first-attempt mastery	Mastered objectives / assessed objectives	objective-mapped tests	Sufficient evidence	Unmapped tests	Assignment, Research	Core
LRN-07 Objective final mastery	Final mastered objectives / assessed objectives	final objective evidence	Sufficient evidence	Unmapped objectives	Assignment, Research	Core
LRN-08 Objective mastery gain	Final mastery% - first mastery%	LRN-06 and LRN-07	Comparable objective definitions	Version changes affecting mappings	Research	Core

G. Built-in hints and AI tutor use

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
TUT-01 Built-in hint requests	Count delivered non-AI hints	hint_interactions	Successfully delivered hints	Failed requests and duplicate clicks	Student, Research	Core
TUT-02 AI interaction count	Count valid AI requests	ai_interactions	Recorded valid questions/hints	Provider retries and failures	Student, Research	Core
TUT-03 AI uptake rate	100 x AI users / AI-enabled starters	condition and AI interactions	TDD+AI condition	TDD-only students	Research	Core
TUT-04 Hint level reached	Highest progressive hint level delivered	hint_level	Delivered hints	Failed deliveries	Student	Core
TUT-05 Average hint level	Mean delivered hint level	hint interactions	Delivered hints	Invalid requests	Assignment	Secondary
TUT-06 Hint escalation rate	Students requesting higher level / hint users	hint sequence	Ordered hint interactions	Repeated same-level requests	Research	Secondary
TUT-07 Post-hint score change	Next score% - pre-hint score%	hint and adjacent attempts	Next meaningful attempt exists	Infrastructure failures	Student, Research	Core
TUT-08 Post-AI score change	Next score% - pre-AI score%	AI and adjacent attempts	Next meaningful attempt exists	No subsequent attempt	Research	Core
TUT-09 Post-hint test improvement	Change in passed tests after hint	test results and hints	Same assignment version	Changed tests	Student, Research	Core
TUT-10 Post-AI test improvement	Change in passed tests after AI response	test results and AI interactions	Same assignment version	Changed tests	Research	Core
TUT-11 Question response rate	Successful AI responses / valid questions	AI request status	Student-generated questions	Automatic retries	AI QA, Operations	Core
TUT-12 AI response latency	responded_at - requested_at	AI timestamps	Completed responses	Cancelled requests	AI QA, Operations	Core
TUT-13 AI response-type distribution	Responses by type / all delivered responses	response_type	Delivered feedback	Provider errors	AI QA	Secondary
TUT-14 AI helpfulness rating	Mean and median optional rating	survey response	Submitted ratings	Missing ratings remain missing	AI QA, Research	Secondary
TUT-15 Question-resolution rate	Questions marked resolved / rated questions	student follow-up	Explicitly rated questions	Missing responses	AI QA	Later
TUT-16 Independent success rate	Successful students with no tutor use / starters	attempts and interactions	Relevant condition	Tutor use after completion	Research	Secondary
TUT-17 Tutor dependency indicator	Tutor interactions per successful assignment	tutor events and completion	Successful students	Failed deliveries	Research	Later
TUT-18 Condition adherence rate	Students following assigned availability / eligible	condition and access events	Eligible participants	Approved deviations	Research	Core

H. AI feedback quality and safety

These metrics normally require instructor review of a protocol-defined sample.

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
AIQ-01 Feedback correctness	Mean reviewer score using fixed rubric	ai_reviews.correctness_rating	Protocol-defined reviewed sample	Unreviewed interactions	AI QA	Core
AIQ-02 Feedback relevance	Mean relevance rating	relevance_rating	Reviewed responses	Unreviewed responses	AI QA	Core
AIQ-03 Feedback clarity	Mean clarity rating	review rubric	Reviewed responses	Missing ratings	AI QA	Secondary
AIQ-04 Debug-step usefulness	Mean debugging-guidance rating	review rubric and response type	Debugging responses	Other response types	AI QA	Secondary

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
AIQ-05 Progressive-hint compliance	Compliant hints / reviewed hints	review rubric and hint level	Reviewed hint responses	Non-hint responses	AI QA	Core
AIQ-06 Direct-solution leakage rate	Solution-leaking responses / reviewed responses	leakage_flag	Reviewed responses	None	AI QA	Core
AIQ-07 Hidden-test leakage rate	Hidden-test leaks / reviewed responses	leakage classification	Reviewed responses	None	AI QA	Core
AIQ-08 Unsupported-claim rate	Incorrect or ungrounded responses / reviewed	correctness review	Reviewed responses	Disputed cases documented	AI QA	Core
AIQ-09 Safe-refusal correctness	Correct refusals / prohibited requests	request type and review	Solution-seeking requests	Normal tutoring requests	AI QA	Core
AIQ-10 False-refusal rate	Unnecessary refusals / allowed requests	review classification	Allowed questions	Prohibited requests	AI QA	Secondary
AIQ-11 Context-use accuracy	Correct use of grader evidence / reviewed responses	review rubric	Responses given grader context	General questions without evidence	AI QA	Core
AIQ-12 Attempt-comparison accuracy	Correct comparisons / reviewed comparison responses	attempts and review	Responses comparing attempts	Responses without comparison	AI QA	Core

I. Assessments and student perception

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
ASM-01 Pre-test score	100 x pre-test score / max	assessment_type, score, max_score	Valid completed pre-test	Practice tests	Research	Core
ASM-02 Post-test score	100 x post-test score / max	assessment fields	Valid completed post-test	Practice tests	Research	Core
ASM-03 Learning gain	Post-test% - pre-test%	ASM-01 and ASM-02	Both assessments complete	Missing assessment pair	Research	Core
ASM-04 Normalized learning gain	(post - pre) / (100 - pre)	pre/post percentages	Pre-test below 100	Denominator zero	Research	Secondary
ASM-05 Retention-test score	100 x retention score / max	assessment_type	Valid delayed test	Immediate post-test	Research	Later
ASM-06 Retention change	Retention% - post-test%	post and retention scores	Both scores available	Missing assessments	Research	Later
SUR-01 Perceived difficulty	Median fixed-scale response	survey response	Submitted response	Missing response	Assignment, Research	Core
SUR-02 Pre-task confidence	Median pre-task confidence	survey response and time	Submitted before assignment	Responses after starting	Research	Secondary
SUR-03 Post-task confidence	Median post-task confidence	survey response and time	Submitted after assignment	Premature responses	Research	Secondary
SUR-04 Confidence gain	Post-confidence - pre-confidence	paired survey responses	Both responses exist	Unpaired responses	Research	Secondary
SUR-05 Cognitive-load rating	Median validated-scale score	survey items	Complete required items	Partial invalid survey	Research	Secondary
SUR-06 Tutor satisfaction	Median satisfaction rating	survey response	AI users	TDD-only students	AI QA, Research	Secondary
SUR-07 Perceived hint usefulness	Median hint-usefulness rating	survey response	Hint users	Non-users	Research	Secondary

J. Assignment and curriculum quality

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
ASG-01 Median final score	Median final score percentage	final attempts	Valid final attempts	Missing attempts	Assignment	Core
ASG-02 Score distribution	Count and percentage by score band	final scores	Valid final attempts	Missing scores shown separately	Assignment	Core
ASG-03 Median attempts to success	Median among successful students	attempts to first success	Successful students	Non-completers separate	Assignment	Core
ASG-04 Median active completion time	Median active time to first success	time metrics	Successful students	Non-completers	Assignment	Core
ASG-05 Most-failed tests	Rank tests by distinct students failing	test_id and outcomes	Valid executed tests	Infrastructure-skipped tests	Assignment	Core
ASG-06 Most-common errors	Rank error categories by affected students	error events	Student-caused errors	Infrastructure errors	Assignment	Core
ASG-07 Most-common misconceptions	Rank by affected students	misconception evidence	Rule-based evidence	Unverified AI labels	Assignment	Core
ASG-08 Test discrimination indicator	Pass-rate difference between high and low performers	test and overall performance	Adequate sample size	Very small classes	Research	Later
ASG-09 Assignment difficulty indicator	Combined completion, score, attempts, and time	assignment metrics	Same assignment version	Mixed versions	Assignment	Secondary
ASG-10 Under-specification signal	High failure + diverse implementations + review flag	failures, feedback, review	Instructor-reviewed cases	Fully automatic conclusions	Assignment	Secondary
ASG-11 Version comparability	Equivalent objectives, tests, and scoring: yes/no	assignment version metadata	Formally mapped versions	Unmapped versions	Research	Core
ASG-12 Publication validation status	Pass/Fail of automated assignment checks	publication test results	Exact published version	Older versions	Instructor	Core

K. Experimental-group comparisons

Group differences support causal claims only when study design and analysis justify that interpretation.

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
CMP-01 Mean outcome difference	Mean TDD+AI - mean TDD-only	outcome and condition	Analysis population	Protocol exclusions	Research	Core
CMP-02 Median outcome difference	Median TDD+AI - median TDD-only	outcome and condition	Analysis population	Missing outcomes handled explicitly	Research	Core
CMP-03 Completion-rate difference	AI completion% - TDD-only completion%	condition and completion	Predefined denominator	Do not change denominator post hoc	Research	Core
CMP-04 Completion risk ratio	AI completion proportion / TDD-only proportion	completion and condition	Valid denominator in both groups	Zero denominator	Thesis export	Secondary
CMP-05 Standardized mean difference	Mean difference / pooled standard deviation	outcome and condition	Appropriate continuous outcome	Tiny samples without caveat	Thesis export	Secondary
CMP-06 Time-to-success difference	Median active-time difference	time and condition	Successful participants	Non-completers analyzed separately	Research	Core
CMP-07 Attempts-to-success difference	Median attempt difference	attempt metric and condition	Successful participants	Non-completers separately	Research	Core
CMP-08 Error-resolution difference	Median resolution difference	error-resolution metrics	Comparable errors	Unresolved errors need censoring	Research	Secondary

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
CMP-09 Learning-gain difference	Mean or median gain difference	pre/post gain and condition	Paired pre/post data	Missing pairs	Research	Core
CMP-10 AI treatment uptake	AI-condition users / AI-condition starters	condition and AI events	TDD+AI participants	TDD-only participants	Research	Core
CMP-11 Intention-to-treat result	Compare by assigned condition	assigned condition and outcomes	All eligible assigned participants	Only predefined exclusions	Thesis export	Core
CMP-12 Per-protocol result	Compare participants meeting adherence rule	adherence and outcomes	Predefined adherence threshold	Post-hoc threshold	Thesis export	Secondary
CMP-13 Adjusted treatment estimate	Predefined regression or ANCOVA estimate	outcome, condition, pre-test, cohort	Model specified before analysis	Post-hoc variable selection	Thesis analysis	Later
CMP-14 Effect confidence interval	Confidence interval around group effect	group outcome data	Appropriate model and assumptions	Informal dashboard-only estimates	Thesis analysis	Core
CMP-15 Assignment-condition interaction	Difference in treatment effect across assignments	condition, assignment, outcome	Adequate repeated observations	Small samples without caveat	Thesis analysis	Later

L. System reliability, security, and data quality

Metric ID and name	Formula	Source fields	Inclusion rules	Exclusion rules	Dashboard	Priority
SYS-01 Grader success rate	Successful grader completions / grader requests	system events	Valid grading requests	Unauthorized requests separately	Operations	Core
SYS-02 Grader failure rate	Infrastructure failures / grader requests	service and status	Confirmed infrastructure failures	Student-code failures	Operations	Core
SYS-03 Median execution latency	Median completed grading duration	duration_ms and status	Completed grading requests	Cancelled requests	Operations	Core
SYS-04 P95 execution latency	95th percentile grading duration	duration_ms	Completed requests	Invalid requests	Operations	Secondary
SYS-05 AI availability rate	Successful or handled AI requests / eligible requests	AI status	AI-enabled requests	AI intentionally disabled	Operations	Core
SYS-06 AI provider failure rate	Provider failures / provider requests	AI and system status	Requests sent to provider	Local validation rejection	Operations	Core
SYS-07 Authentication failure rate	Failed logins / login attempts	authentication events	Real login requests	Tagged QA automation	Operations	Secondary
SYS-08 Unauthorized-access block rate	Blocked unauthorized requests / unauthorized attempts	security events	Confirmed unauthorized requests	Normal validation errors	Operations	Core
SYS-09 Missing-event rate	Missing required chains / expected chains	IDs across entities	Valid workflow requests	Cancelled pre-action events	Data QA	Core
SYS-10 Duplicate-event rate	Duplicate logical events / recorded events	request and event IDs	Study-window events	Legitimate repeated actions	Data QA	Core
SYS-11 Orphan-record count	Records missing required parent	foreign keys and entity IDs	All study records	Intentional anonymization	Data QA	Core
SYS-12 Version-missing rate	Attempts without assignment_version / attempts	attempt version field	All valid attempts	Marked legacy data	Data QA	Core
SYS-13 Condition-missing rate	Study attempts without condition / study attempts	attempt and participant condition	Study participants	Non-study demo sessions	Data QA	Core
SYS-14 Timestamp-order error rate	Impossible event chains / event chains	event timestamps	Complete chains	Documented clock migrations	Data QA	Core
SYS-15 Export reconciliation	Exported matching rows / expected rows	export and database counts	Completed export	Explicit private-field exclusions	Data QA	Core
SYS-16 Backup verification rate	Verified backups / scheduled backups	backup records	Scheduled windows	Cancelled maintenance	Operations	Core
SYS-17 Restore-test success	Successful restore rehearsals / attempted rehearsals	restore-test logs	Controlled tests	Production incidents separately	Operations	Core
SYS-18 Demo-data contamination count	Demo records incorrectly entering research data	demo flag and exports	Research datasets	Correctly excluded demo activity	Data QA	Core

4. Recommended v5.2 core collection set

The complete dictionary is a decision catalog. The first implementation should collect a smaller dependable core rather than every possible measure.

Term	Formal definition
Population	POP-01 through POP-07, plus POP-10
Outcomes	OUT-01 through OUT-13, plus OUT-15
Attempts	ATT-01 through ATT-06, ATT-09, ATT-10
Time	TIME-01, TIME-03, TIME-05 through TIME-09
Errors and TDD	ERR-01 through ERR-15
Learning	LRN-01 through LRN-08
Tutor	TUT-01 through TUT-12, plus TUT-18
AI safety	AIQ-01, AIQ-02, AIQ-05 through AIQ-09, AIQ-11
Assessments	ASM-01 through ASM-03, plus SUR-01
Assignment quality	ASG-01 through ASG-07, ASG-11, ASG-12
Comparisons	CMP-01 through CMP-03, CMP-06, CMP-07, CMP-09 through CMP-11
Reliability	SYS-01 through SYS-03, SYS-05, SYS-06, SYS-09 through SYS-18

5. Decisions required before implementation

1. Define the exact completion rule for every assignment version.

2. Approve the idle threshold; 15 minutes is the proposed default.

3. Retain both final-attempt and best-attempt scores; never silently substitute one for the other.

4. Define misconception-detection rules with instructors.

5. Map every test to one or more learning objectives.

6. Define TDD_ONLY and TDD_AI conditions and prevent students from changing them.

7. Freeze inclusion, withdrawal, late-work, and missing-data rules before enrollment.

8. Approve an instructor AI-review rubric and sampling protocol.

9. Decide which metrics students may see and which remain instructor/research-only.

10. Freeze and version this dictionary before collecting real research data.

6. Research integrity rules

Term	Formal definition
Student versus system failure	Never classify an infrastructure error as a student failure or failed learning attempt.
Time measurement	Use active time, not total browser-open time.
Assignment comparability	Compare outcomes only within the same version or formally mapped equivalent versions.
AI role	AI provides feedback only. Deterministic tests remain authoritative for grading.
Privacy	Use participant codes in research views and exports; exclude credentials, source code, hidden tests, and solutions.
Causality	Post-feedback improvement is an association unless the study design supports a causal conclusion.
Reproducibility	Keep raw append-only evidence and version formulas, status values, assignment definitions, and study configuration.
Missing data	Report missingness explicitly. Do not silently convert missing values to zero.